

The Fibonacci Sequence and Golden Ratio

A detailed study of the Fibonacci sequence, its properties, and its connection to the Golden Ratio

Eklavya Raman
er24ms024@iiserkol.ac.in

December 7, 2025

I Exercises after Tutorials

Exercise I: Prove or disprove that the rank of a Gram Matrix is invariant under a change of basis.

Proof. We claim that the rank of a Gram matrix is invariant under a change of basis. Let G be the Gram matrix in the original basis and G' be the Gram matrix in the new basis. Let $B(u, v)$ be the form associated with the Gram matrix G . The Gram matrix G' in the new basis is given by $G' = P^T G P$, where P is the change of basis matrix. We show that $\dim \text{Im}(B) = \dim \text{Im}(B')$. We have $\dim \ker(B)$ is the dimension of the kernel of the form B . Suppose $u, v \in \ker(B)$, then $B(u, v) = 0$. Under the change of basis, we have $u' = P^T u$, where $u' = P^T u$ and $v' = P^T v$. Then, $B'(u', v') = B(P^T u, P^T v) = P^T B(u, v) P = 0$. Thus, $u', v' \in \ker(B')$. The converse is also true. If $u', v' \in \ker(B')$, then $B'(u', v') = 0$. This implies that $B(Pu', Pv') = 0$, which means $u, v \in \ker(B)$. Therefore, we have $\dim \ker(B) = \dim \ker(B')$. By rank-nullity theorem, we have:

$$\dim \text{Im}(B) + \dim \ker(B) = \dim \text{Im}(B') + \dim \ker(B')$$

Combining these results, we find that:

$$\dim \text{Im}(B) = \dim \text{Im}(B')$$

This contradicts our assumption that the rank of the Gram matrix changes under a change of basis. Therefore, the rank of a Gram matrix is invariant under a change of basis. $\square$

Exercise II: Let V be a finite-dimensional real vector space and let $W_1, W_2, \dots, W_n$ be proper subspaces of V . Show that $V \neq \bigcup_{i=1}^n W_i$.

Proof. We will prove this by induction on n , the number of proper subspaces.

Base Case: For $n = 1$, let W_1 be a proper subspace of V . Since W_1 is a proper subspace, there exists at least one vector $v \in V$ such that $v \notin W_1$. Therefore, $V \neq W_1$.

Inductive Step: Assume that for some $k \geq 1$, the statement holds for $n = k$; that is, $V \neq \bigcup_{i=1}^k W_i$. We need to show that it also holds for $n = k + 1$.

Consider the subspaces $W_1, W_2, \dots, W_k, W_{k+1}$. By the inductive hypothesis, there exists a vector $v \in V$ such that $v \notin \bigcup_{i=1}^k W_i$. Now, we need to check if this vector v is in W_{k+1} .

If $v \notin W_{k+1}$, then we have found a vector in V that is not in any of the subspaces $W_1, W_2, \dots, W_k, W_{k+1}$, and thus $V \neq \bigcup_{i=1}^{k+1} W_i$.

If $v \in W_{k+1}$, we can choose another vector $u \in V$ such that $u \notin W_{k+1}$ (since W_{k+1} is a proper subspace). Now consider the vector $w = v + u$. Since both v and u are in V , their sum w is also in V .

If $w \in \bigcup_{i=1}^{k+1} W_i$, then it must belong to at least one of the subspaces. However, since $v \notin \bigcup_{i=1}^k W_i$, w cannot be in any of the first k subspaces. Therefore, w must be in W_{k+1} . But this contradicts our choice of u since $w = v + u$ and $v \in W_{k+1}$ implies that u must also be in W_{k+1} , which is a contradiction.

Thus, in either case, we can find a vector in V that is not in any of the subspaces $W_1, W_2, \dots, W_{k+1}$. Therefore, by induction, we conclude that for any finite number of proper subspaces $W_1, W_2, \dots, W_n$ of V , we have $V \neq \bigcup_{i=1}^n W_i$. $\square$

Exercise III: We need to show $\text{rank}(M) = \text{rank}(M^T M)$ for any real matrix $M \in M_n(F)$.

Proof. Consider the transformation $T : F^n \rightarrow F^n$ defined by $T(x) = Mx$. The rank of the matrix M is equal to the dimension of the image of the transformation T , i.e., $\text{rank}(M) = \dim(\text{Im}(T))$.

Now, consider the transformation $S : F^n \rightarrow F^n$ defined by $S(x) = M^T x$. We show that $\dim(\text{Im}(T)) = \dim(\text{Im}(S \cdot T))$.

Suppose $x \in \ker T$, $x \neq 0$, then $S(T(x)) = S(0) = 0$. Thus, $x \in \ker(S)$. Therefore, $\ker(T) \subseteq \ker(S \cdot T)$.

Suppose $x \in \ker(S \cdot T)$, $x \neq 0$, then $S(T(x)) = 0$. Therefore -

$$x^T M^T M x = 0 \implies (Mx)^T (Mx) = 0 \implies \|Mx\|^2 = 0 \implies x \in \ker(T)$$

Thus, $\ker(S \cdot T) \subseteq \ker(T)$.

Combining both results, we have $\ker(T) = \ker(S \cdot T)$.

By rank-nullity theorem, we have:

$$\dim(\text{Im}(T)) = \dim(\text{Im}(S \cdot T))$$

Hence proved. $\square$

Exercise IV: $\mathbb{B} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$ is a basis for $M_2(\mathbb{R})$. We have $\mathbb{B}' = \{e_1, e_2, e_3, e_4\}$ is the standard basis for $M_2(\mathbb{R})$. Find the change of basis matrix from $\mathbb{B}$ to $\mathbb{B}'$. We have the following representations of the basis vectors in $\mathbb{B}$ in terms of the standard basis $\mathbb{B}'$:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1 \cdot e_1 + 0 \cdot e_2 + 0 \cdot e_3 + 1 \cdot e_4$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 1 \cdot e_1 + 0 \cdot e_2 + 0 \cdot e_3 - 1 \cdot e_4$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = 0 \cdot e_1 + 1 \cdot e_2 - 1 \cdot e_3 + 0 \cdot e_4$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = 0 \cdot e_1 + 1 \cdot e_2 + 1 \cdot e_3 + 0 \cdot e_4$$

Thus, the change of basis matrix from $\mathbb{B}$ to $\mathbb{B}'$ is given by:

$$P = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \end{pmatrix}$$

Exercise V: We have $W_1, W_2, \dots, W_k$ are subspaces of a finite-dimensional vector space V such that

$$W_1 + W_2 + \dots + W_k = V$$

and

$$W_i \cap (W_1 + W_2 + \dots + W_{i-1}) = \{0\}$$

for each i . Show that V is the direct sum of $W_1, W_2, \dots, W_k$.

Proof. We need to show that every vector $v \in V$ can be uniquely expressed as a sum of vectors from each subspace W_i .

Since $W_1 + W_2 + \dots + W_k = V$, for any vector $v \in V$, there exist vectors $w_1 \in W_1, w_2 \in W_2, \dots, w_k \in W_k$ such that:

$$v = w_1 + w_2 + \dots + w_k$$

Now, we need to show that this representation is unique. Suppose there exists another representation of v :

$$v = w'_1 + w'_2 + \dots + w'_k$$

where $w'_i \in W_i$ for each i . Subtracting the two representations, we get:

$$0 = (w_1 - w'_1) + (w_2 - w'_2) + \dots + (w_k - w'_k)$$

Let $u_i = w_i - w'_i$. Then we have:

$$0 = u_1 + u_2 + \dots + u_k$$

where $u_i \in W_i$ for each i . We will show that each $u_i = 0$.

Consider the last term u_k . Since $u_1 + u_2 + \dots + u_k = 0$, we can rearrange this to:

$$u_k = -(u_1 + u_2 + \dots + u_{k-1})$$

The right-hand side belongs to the subspace $W_1 + W_2 + \dots + W_{k-1}$. However, by the given condition, we have:

$$W_k \cap (W_1 + W_2 + \dots + W_{k-1}) = \{0\}$$

This implies that the only way for u_k to equal the right-hand side is if $u_k = 0$.

Repeating this argument for each u_i , we find that:

$$u_i = 0 \quad \text{for all } i = 1, 2, \dots, k$$

This shows that the representation of v as a sum of vectors from each subspace W_i is unique.

Hence, V is the direct sum of $W_1, W_2, \dots, W_k$. □